Digital Egypt Pioneers Initiative (DEPI) – Phase 5

Graduation Project – Project Documentation

ShopSense

E-Commerce Customer Intelligence Platform

Customer Value & Purchase Prediction

Part 1: Project Planning & Management

Part 2: Literature Review

Part 3: Requirements Gathering

Item	Details
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Submission Date	19 October 2026

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Part 1: Project Planning & Management

1.1 Project Overview

ShopSense is a database-backed analytics and machine-learning system that gives an e-commerce company a single, integrated view of its customers, orders, products and payments, and uses that view to predict future customer behaviour. The platform covers the full data lifecycle: a relational database, SQL analytics, Python-based cleaning, exploratory analysis and feature engineering, a machine-learning model, and an interactive dashboard that presents the results to business users.

Problem statement

E-commerce companies collect large volumes of customer and transaction data, but this data is often scattered across separate sources and used only for basic operational reporting. As a result, management lacks a consolidated view of purchasing behaviour (who buys, how often, how much, and what) and cannot anticipate which customers are likely to return. Without this insight, marketing spend is untargeted, loyal customers are not recognised, and one-time buyers are not identified early enough to be re-engaged.

Proposed solution

The team will integrate customer, order, product, category and payment data into one relational database, analyse purchasing patterns with SQL and Python, and train a machine-learning model for repeat-purchase prediction: given a customer's first-order behaviour, predict whether the customer will place another order within a defined follow-up window (initially 90 days). Results (revenue, customers, products, segments and model predictions) are exposed through a dashboard.

1.2 Project Objectives

General objective

Build a complete E-Commerce Customer Intelligence Platform that identifies purchasing patterns and predicts a defined customer outcome (repeat purchase).

Specific objectives

- Design and implement a normalised relational database containing Customers, Orders, Products, Categories and Payments.
- Produce SQL analytics on customer purchase frequency, revenue, average order value (AOV), product performance and repeat-purchase patterns.
- Clean the data, treat missing values, engineer behavioural features (e.g., RFM, basket size, payment type, delivery time) and perform EDA and correlation analysis in Python.
- Train, tune and evaluate machine-learning classifiers for repeat-purchase prediction and select the best-performing model using appropriate metrics for imbalanced data.
- Segment customers (e.g., RFM-based clusters) to support targeted marketing decisions.
- Deliver an interactive dashboard covering revenue, customers, products, customer segments and model predictions.

- (Optional extension) Track experiments with MLflow and expose the prediction model through a FastAPI endpoint.

1.3 Project Scope

In scope

- Relational database layer with five core entities plus an order-items link table.
- SQL analytics queries and views supporting the business questions listed above.
- Python data pipeline: cleaning, missing-value treatment, feature creation, EDA and correlation analysis.
- One machine-learning task: repeat-purchase prediction (binary classification), including baseline, model comparison, tuning and evaluation.
- Customer segmentation to enrich the dashboard.
- Interactive dashboard with five views: Revenue, Customers, Products, Customer Segments and Model Predictions.
- Project documentation, GitHub repository, and a final demonstration.

Out of scope

- Real-time data streaming and live integration with a production e-commerce website.
- Building a customer-facing web shop or a recommendation engine.
- Deep-learning or NLP models (e.g., on review text).
- Cloud deployment at production scale (the optional FastAPI service runs locally or in a container).

Data source (proposed)

The team plans to use the public Brazilian E-Commerce Dataset by Olist [10], which contains about 100,000 anonymised orders from 2016 to 2018 spread over several related tables (customers, orders, order items, payments, products, sellers, reviews and geolocation). Its structure maps directly onto the required entities. In this dataset customer_id is generated per order, whereas customer_unique_id identifies the same real customer across orders; all customer-level metrics (purchase frequency, repeat customers, RFM and the ML label) will therefore be computed on customer_unique_id. The final dataset choice will be confirmed with the Technical Instructor.

1.4 Technology Stack

Layer	Technologies	Purpose
Database	PostgreSQL (or MySQL), SQL	Store and relate customers, orders, products, categories and payments; run analytical queries
Data analysis	Python, pandas, NumPy, Matplotlib, Seaborn	Cleaning, missing-value treatment, feature creation, EDA, correlation analysis
Machine learning	scikit-learn, XGBoost / LightGBM, imbalanced-learn	Baseline and advanced classifiers, class-imbalance handling, evaluation

Layer	Technologies	Purpose
Dashboard	Power BI or Streamlit (to be finalised)	Interactive visualisation of KPIs, segments and predictions
MLOps (optional)	MLflow, FastAPI	Experiment tracking and model serving through a REST API
Collaboration	GitHub, Google Drive, Jupyter / VS Code	Version control, shared documents, development environment

1.5 Methodology and System Architecture

The team follows an iterative, CRISP-DM-inspired workflow adapted to a four-week schedule: business understanding, data understanding, data preparation, modelling, evaluation and deployment (dashboard and optional API). Work is organised in weekly milestones, with a short review at the end of each week and a task board in GitHub to track progress.

Stage	Description	Output
1. Data ingestion	Load raw CSV files into staging tables in the database	Populated relational database
2. SQL analytics	Run aggregate queries and create views for KPIs	Query scripts and result tables
3. Python preparation	Extract data from the database, clean it, treat missing values and engineer features	Clean modelling dataset
4. EDA and correlation	Explore distributions, trends and relationships between variables	EDA notebook with charts
5. Machine learning	Train, tune and evaluate repeat-purchase classifiers; segment customers	Selected model and metrics
6. Dashboard	Connect the dashboard to the analytics tables and model outputs	Five-view interactive dashboard
7. Serving (optional)	Log experiments in MLflow and serve predictions via FastAPI	Tracked runs and prediction endpoint

1.6 Project Timeline

The implementation phase is planned over four weeks in November 2026 (1–28 November), with 29–30 November reserved as a buffer for final fixes and submission, so that all deliverables are complete before December. The Project Documentation (Parts 1–3) is submitted on 19 October 2026.

Task	W1 1–7 Nov	W2 8–14 Nov	W3 15–21 Nov	W4 22–28 Nov	Owner
Finalise requirements and acquire data	■				Moaaz, all
Database design and implementation	■				Marwan
SQL analytics	■	■			Marwan
Data cleaning and missing values		■			Belal
Feature engineering, EDA, correlation		■			Belal
ML modelling and evaluation		■	■		Yahya, Moaaz
Customer segmentation			■		Yahya
Dashboard development			■	■	Mariam
MLflow and FastAPI (optional)				■	Moaaz
Integration, testing, bug fixing				■	All
Final documentation, demo, submission				■	Mariam, Moaaz

Table 1.1: Four-week project plan (W = week). Shaded cells show the weeks in which each task is active.

Milestone	Target date	Deliverable
M0: Documentation (Parts 1–3)	19 Oct 2026	PDF uploaded to the GitHub repository and submitted on the LMS
M1: Database ready	7 Nov 2026	Schema, loaded data and first SQL queries
M2: Clean dataset and EDA	14 Nov 2026	Feature-ready dataset, EDA notebook, SQL analytics
M3: Model and dashboard draft	21 Nov 2026	Trained model with metrics; first dashboard version
M4: Final platform	28 Nov 2026	Integrated platform, optional MLflow/FastAPI, final report and demo

1.7 Team Structure, Tasks and Responsibilities

The team has five members. The role distribution below is provisional and will be reviewed after the first team meeting; the Team Leader also contributes technically. Every member participates in testing and in the final documentation.

Member	Role	Main responsibilities
Moaaz Mohamed Ibrahim	Team Leader / ML & MLOps	Planning, coordination and instructor communication ML pipeline integration; MLflow and FastAPI (optional) Repository management and final review
Marwan Adel Ali	Database & SQL Developer	Schema design, data loading and constraints SQL analytics queries and views
Belal Mahmoud Fathy	Data Analyst / Data Preparation	Cleaning and missing-value treatment Feature creation, EDA and correlation analysis
Yahya Mohamed Abdelhalim	ML Engineer	Model training, tuning and evaluation Customer segmentation (RFM and clustering)
Mariam Mohamed Hany	Dashboard Developer / Documentation	Dashboard design and implementation Documentation, QA and presentation material

Working agreements: weekly team meeting, shared task board, feature branches with pull-request review on GitHub, and a single repository for the whole team as required by the programme.

1.8 Risk Management

#	Risk	Likelihood	Impact	Mitigation
R1	Class imbalance: few customers place a second order, so the model favours the majority class	High	High	Use stratified splits, class weights or SMOTE, and evaluate with recall, F1 and PR-AUC rather than accuracy alone
R2	Data quality problems (missing values, duplicates, inconsistent types)	High	Medium	Documented cleaning rules, validation checks, and keeping the raw data untouched
R3	Data leakage in the repeat-purchase label or features	Medium	High	Define the observation and outcome windows up front; build features only from data before the cut-off date
R4	Schedule delay because of study or exam workload	Medium	High	Weekly checkpoints, early start on critical tasks, and a buffer at the end of November
R5	Dependency between tasks (e.g., ML cannot start before clean data)	Medium	Medium	Share a sample dataset early and agree on column names and data types
R6	Low model performance	Medium	Medium	Start with a simple baseline, iterate on features, compare several algorithms, and report honestly with a clear baseline
R7	Tool or environment issues (versions, database setup, dashboard licence)	Medium	Medium	Shared requirements file, documented setup steps, and an open-source fallback (Streamlit)
R8	Merge conflicts or lost work in GitHub	Low	Medium	Branching strategy, regular commits and pull-request reviews
R9	Team member unavailable	Low	High	Pair tasks, document work in the repository, and keep roles cross-trained

#	Risk	Likelihood	Impact	Mitigation
R10	Optional extension (MLflow/FastAPI) consumes time needed for core scope	Medium	Low	Treat as a stretch goal started only after M3 is met

1.9 Success Criteria and Key Performance Indicators

Area	Indicator	Target
Delivery	All mandatory deliverables completed by the end of November 2026	100% of core scope
Database	Five core entities implemented with primary and foreign keys and no orphan records	0 integrity violations
SQL analytics	Required analyses (frequency, revenue, AOV, product performance, repeat patterns) implemented and documented	5 of 5 analyses
Data quality	Remaining missing values in the modelling dataset after treatment	0 in model features
Model	ROC-AUC on the held-out test set for repeat-purchase prediction	≥ 0.70 and better than the baseline
Model	Recall and F1-score for the repeat-buyer (minority) class reported and interpreted	Reported for all candidate models
Dashboard	Required views available (Revenue, Customers, Products, Segments, Predictions)	5 of 5 views
Reproducibility	Project can be re-run from the repository using the README instructions	Verified by a team member who did not write the code
Optional	MLflow runs logged and FastAPI /predict endpoint working	Achieved if time allows

Part 2: Literature Review

2.1 Purpose and Approach

This review examines existing research, public projects and common practices related to ShopSense in three areas: (1) predicting repeat purchases and repeat buyers, (2) customer segmentation with RFM analysis and clustering, and (3) public projects built on e-commerce transaction data, in particular the Olist dataset. For each source we note what it contributes, its limitations relative to our project, and what we adopt. Sources were identified through web searches of academic repositories, journals and GitHub in October 2026; all are listed in Section 2.7. Reported figures are quoted as stated by the original authors and were obtained on their own datasets, so they are not directly comparable with our future results.

2.2 Repeat-Purchase and Repeat-Buyer Prediction

Zhang and Wang [1] predict repurchase behaviour using Alibaba mobile e-commerce data. They build a feature set made of user, product and user–product interaction features and propose an improved deep forest model; they report that adding interaction features improves accuracy and that the model also reduces training time compared with existing models. The paper shows how much the quality of feature engineering matters, which supports our plan to invest heavily in behavioural features.

Zhang et al. [2], working with Tmall data, observe that the share of repeat buyers in e-commerce is low (they cite a repurchase rate of only 6.1% for Tmall), builds features through manual construction and algorithmic selection, and applies SMOTE to handle class imbalance. It compares logistic regression and factorization machines with XGBoost and LightGBM, and finally builds a two-layer stacking model. This directly informs two of our design decisions: treating class imbalance explicitly and comparing a simple baseline with gradient-boosting models.

Usman [3] targets one-time promotion buyers who never return, using XGBoost, LightGBM and histogram-based gradient boosting with feature engineering, and reports a test accuracy of about 96% for the best model. Because the share of repeaters is small, accuracy alone can be misleading in this setting, which is why we plan to report recall, F1 and PR-AUC. Sun [4] combines factor analysis with K-means-based processing for repeat-purchase prediction and reports accuracy above 90%, while a 2025 study [5] addresses purchase-frequency prediction with boosting-based and hybrid multi-stage machine-learning approaches and highlights historical behaviour, purchase intervals and engagement as informative features.

2.3 Customer Segmentation with RFM and Clustering

Several studies segment e-commerce customers using Recency, Frequency and Monetary (RFM) values combined with clustering. Serwah et al. [6] combine RFM with weighted K-means and report a higher silhouette score (0.40) than standard K-means (0.30). Wong et al. [7] compare K-means and hierarchical clustering on RFM data and report that hierarchical clustering gave better cluster-quality scores in their experiment. Hilmy et al. [8] go a step further by building a web-based segmentation system with RFM and K-means that visualises results for business users, and report that K-means outperformed K-Medoids on their data. Reddy [9] uses the elbow method with K-means to label groups such as Champions, Loyal and Promising customers.

These works confirm that RFM-based segmentation is a well-established, interpretable approach and that cluster quality should be validated (elbow method, silhouette score). They also show that a dashboard presentation of segments is valuable. We adopt RFM with K-means as our segmentation method and evaluate it with the same metrics.

2.4 Public Projects on E-Commerce Transaction Data

The Olist dataset [10] is a widely used public dataset of about 100,000 anonymised orders from 2016 to 2018 with information on order status, price, payment, freight, customer location, product attributes and reviews. Public repositories built on it include a customer-segmentation and lifetime-value project [11] and a predictive-analytics project for sales and marketing [12]; the latter reports that only about 3.4% of customers made a repeat purchase, which confirms that class imbalance will be a central challenge for repeat-purchase prediction on this data. Many of these projects are notebook-based: they focus on analysis or modelling but do not usually combine a relational database, SQL analytics, a model and a dashboard in one integrated platform.

2.5 Comparison with ShopSense

Source / approach	What it offers	Limitations relative to our scope	What ShopSense adopts or does differently
Zhang & Wang [1] – deep forest repurchase model	Rich user, product and interaction features; accuracy and training-time gains	Model-focused research; no database layer, SQL analytics or dashboard described	Adopt feature groups (customer, product, interaction); add database, SQL and dashboard layers
Tmall repeat-buyer study [2]	Feature selection, SMOTE, boosting and stacking	Focus on modelling performance; complex stacking is heavy for a four-week project	Adopt SMOTE/class weights and the baseline-versus-boosting comparison; skip stacking
Usman [3] – repeat-buyer thesis	Gradient-boosting models with feature engineering and high reported accuracy	Accuracy can be inflated by class imbalance; no end-to-end platform described	Evaluate with recall, F1, PR-AUC and a confusion matrix, not accuracy alone
Sun [4] and purchase-frequency study [5]	Alternative formulations (clustering-based features, purchase frequency)	Different prediction targets; limited reproducible details in the abstracts	Use as background; keep a clearly defined binary repeat-purchase target
RFM + clustering studies [6]–[9]	Interpretable segments and cluster-quality validation; one includes a web-based system [8]	Segmentation only; they do not predict future customer behaviour	Use RFM + K-means for segments and combine them with a predictive model in one dashboard
Olist-based GitHub projects [11], [12]	Practical examples of segmentation, LTV and sales prediction on the same dataset	Typically notebook-only; no relational database or integrated dashboard	Build the full pipeline: database, SQL, Python, ML and dashboard in one repository
General-purpose BI dashboards (e.g., Power BI, Tableau)	Strong reporting and visualisation on historical data	Usually descriptive; predictions and custom features need extra development	Embed model predictions and segments directly in the dashboard

Table 2.1: Comparison of related work with the proposed project.

2.6 Gap Analysis and Lessons Learned

- **Gap:** most reviewed works handle a single task (prediction or segmentation) and rarely combine database, analytics, model and dashboard in one deliverable. ShopSense integrates all of them.
- **Lesson 1:** feature engineering is a major driver of repeat-purchase model quality [1], [2], [3]; we dedicate a full phase to it.
- **Lesson 2:** repeat buyers are a small minority [2], [12]; we use stratified splitting, imbalance handling and metrics suited to imbalanced classes.
- **Lesson 3:** simple, interpretable models should be the baseline before more complex ones; we start with logistic regression and compare with tree-based boosting.
- **Lesson 4:** RFM + K-means is a proven, explainable way to create segments [6]–[9]; clusters must be validated with the elbow method and silhouette score.
- **Lesson 5:** business users need results presented visually [8]; the dashboard is a first-class deliverable, not an afterthought.

2.7 References

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Part 3: Requirements Gathering

3.1 Stakeholders and Their Needs

Requirements were derived from the project brief, the problem statement and typical needs of an e-commerce business. Stakeholders are listed below with the main needs the platform must satisfy. Because this is a graduation project, business roles are represented by the instructor's brief and the team's analysis of the domain.

Stakeholder	Role	Main needs
Business / Management	Strategic decisions	Clear view of revenue, customers and best-selling products Understanding of customer value and loyalty Early view of which customers are likely to return
Marketing team	Campaigns and retention	Customer segments for targeted campaigns List of customers likely or unlikely to repeat-purchase Insight into product and category performance
Sales / Category managers	Product and pricing	Product and category performance, AOV and repeat-purchase patterns Payment-method preferences
Data analysts	Daily analysis	Reliable, documented database and SQL views Clean datasets and reproducible notebooks
Data scientists / ML engineers	Modelling	Well-defined target, leakage-free features, experiment tracking Ability to retrain and compare models
Dashboard users	Consumption of results	Easy-to-read, interactive visuals with filters Consistent numbers across dashboard and SQL
IT / DevOps (optional)	Integration	Simple prediction API with clear input and output format
Technical Instructor / evaluators	Assessment	Complete deliverable covering every required component Clear documentation and a working demo

3.2 Functional Requirements

Priority levels follow the MoSCoW scheme: M = Must have, S = Should have, C = Could have (optional extension).

A. Database layer

ID	Requirement	Priority
FR-01	The system shall store customers, orders, order items, products, categories and payments in a relational database with primary and foreign keys.	M
FR-02	The system shall enforce referential integrity (e.g., each order belongs to an existing customer; each product belongs to a category).	M

ID	Requirement	Priority
FR-03	The system shall support loading the raw source files into the database through a repeatable script.	M
FR-04	The system shall provide SQL views for the main analytical outputs to be reused by Python and the dashboard.	S

B. SQL analytics

ID	Requirement	Priority
FR-05	Calculate customer purchase frequency (number of orders per customer_unique_id) and its distribution.	M
FR-06	Calculate total revenue overall and by period (month), category and payment type.	M
FR-07	Calculate average order value (overall, by segment and by period).	M
FR-08	Rank products and categories by sales volume, revenue and order count.	M
FR-09	Identify repeat-purchase patterns by customer_unique_id: share of repeat customers, time between orders and first-order characteristics.	M

C. Python analytics

ID	Requirement	Priority
FR-10	Clean the data: remove duplicates, fix data types, handle inconsistent values and outliers, and document each rule.	M
FR-11	Detect and treat missing values with justified strategies (deletion, imputation or flagging).	M
FR-12	Create features: for analysis and segmentation, RFM metrics, order count and AOV over the full history; for the ML model, only first-order features available at the prediction point (order value, basket size, product category, payment type and installments, freight, delivery time and delay, review score if submitted before the cut-off), as defined in Section 3.5.	M
FR-13	Perform exploratory data analysis with summary statistics and visualisations.	M
FR-14	Perform correlation analysis among features and with the target variable.	M

D. Machine learning

ID	Requirement	Priority
FR-15	Define the target variable and the eligible customer population exactly as specified in Section 3.5 (second order within 90 days after the prediction point; full follow-up window required; no future information in the features).	M
FR-16	Split the data into training and test sets using a time-based split of first-order cohorts (with stratified cross-validation inside the training set for tuning); fit all preprocessing and resampling on training data only.	M
FR-17	Train a baseline model (e.g., logistic regression) and at least two advanced models (e.g., Random Forest and XGBoost/LightGBM).	M

ID	Requirement	Priority
FR-18	Handle class imbalance (class weights and/or SMOTE) and compare the effect.	M
FR-19	Evaluate models using ROC-AUC, precision, recall, F1-score, PR-AUC and the confusion matrix; select the final model.	M
FR-20	Explain the model through feature importance (and optionally SHAP values).	S
FR-21	Segment customers using RFM and K-means, validated with the elbow method and silhouette score.	S
FR-22	Save the final model and its predictions for each customer so that the dashboard can display them.	M

E. Dashboard

ID	Requirement	Priority
FR-23	Revenue view: total revenue, monthly trend, AOV and revenue by category/payment type.	M
FR-24	Customers view: number of customers, new versus repeat customers, purchase frequency and geography.	M
FR-25	Products view: top products and categories and their performance over time.	M
FR-26	Customer segments view: segment sizes, profiles and revenue contribution.	M
FR-27	Model predictions view: predicted repeat-purchase probability per customer or group, model metrics and feature importance.	M
FR-28	Provide filters (date range, category, segment) that update all relevant visuals.	S

F. Optional extension

ID	Requirement	Priority
FR-29	Log experiments (parameters, metrics, model artefacts) in MLflow for every training run.	C
FR-30	Expose the trained model through a FastAPI endpoint that accepts customer features and returns the repeat-purchase probability, with input validation and interactive API documentation.	C

3.3 Non-Functional (Quality) Requirements

ID	Category	Requirement	Measure / acceptance criterion
NFR-01	Performance	Analytical SQL queries and dashboard pages shall respond quickly on the project dataset.	Typical queries under 5 seconds; dashboard page load under 10 seconds
NFR-02	Performance	Single-customer prediction through the API (if implemented) shall be fast.	Response in under 1 second on a local machine
NFR-03	Security	Database credentials and API keys shall not be stored in source code or committed to GitHub.	Credentials kept in environment variables / .env files excluded by .gitignore

ID	Category	Requirement	Measure / acceptance criterion
NFR-04	Security / privacy	Customer data shall be anonymised and access to the database restricted.	Public anonymised dataset used; no personal identifiers; separate read-only user for the dashboard
NFR-05	Usability	The dashboard shall be understandable to non-technical users.	Clear titles, consistent colours, tooltips, and a maximum of two clicks to reach any view
NFR-06	Reliability / data quality	Data loading and cleaning shall be validated to prevent wrong results.	Row-count and null checks after each load; documented cleaning rules
NFR-07	Reproducibility	Anyone shall be able to rerun the full pipeline.	README with setup steps, requirements.txt, fixed random seeds
NFR-08	Maintainability	Code shall be modular, commented and version-controlled.	Consistent naming, notebooks and scripts organised by stage, pull-request reviews
NFR-09	Scalability	The design shall allow larger datasets or new data to be added.	Normalised schema, indexes on key columns, parameterised scripts
NFR-10	Model quality	The final model shall outperform a naive baseline on the test set.	ROC-AUC ≥ 0.70 and higher than the baseline; metrics reported for the minority class
NFR-11	Portability	The solution shall run on common student hardware using open-source tools.	Runs on a standard laptop with Python and PostgreSQL/MySQL

3.4 Data Requirements

The database will contain the entities below. Attribute names are indicative and will be finalised during schema design. Important: in the Olist data a new customer_id is issued for every order, so the same person can appear under several customer_id values. The database keeps customer_id as the key that preserves the order–customer relationship, and every customer-level analysis (frequency, repeat purchase, segments, ML label) groups records by customer_unique_id.

Table	Main attributes	Relationships
customers	customer_id (PK, one per order record), customer_unique_id (stable customer identifier), city, state, zip_prefix	Each order references a customer_id; customer_id values are grouped by customer_unique_id to obtain all orders of one real customer
orders	order_id (PK), customer_id (FK), order_status, purchase_timestamp, delivered_timestamp, estimated_delivery_date	Belongs to one customer; has many order items and payments
order_items	order_id (FK), order_item_id, product_id (FK), price, freight_value	Links orders and products (many-to-many)
products	product_id (PK), category_id (FK), weight, dimensions, number of photos	Belongs to one category
categories	category_id (PK), category_name (and English translation)	One category has many products

Table	Main attributes	Relationships
payments	order_id (FK), payment_sequential, payment_type, installments, payment_value	Belongs to one order

3.5 Machine-Learning Target Definition and Leakage Prevention

This section fixes how the repeat-purchase label is built, which customers are used, and which information the model may see. Its purpose is to make the prediction realistic: the model must predict only from what is known at the moment of prediction.

Item	Definition
Unit of analysis	One row per real customer, identified by customer_unique_id.
Qualifying first order	The customer's earliest order with status delivered. Cancelled, unavailable and never-delivered orders are excluded from the cohort.
Prediction point (cut-off)	The delivery timestamp of the first order. The model scores a customer at this moment.
Label (target)	repeat_90d = 1 if the customer places a second order whose purchase timestamp falls within 90 days after the prediction point; otherwise 0.
Eligible customers	Only customers whose prediction point is at least 90 days before the last date in the dataset, so every customer has a complete follow-up window. Later customers are excluded from training and testing (they could only be scored, not evaluated).
Allowed features	First-order information known at or before the prediction point: order value, number of items, product category, payment type and installments, freight value, delivery duration and delay versus the estimated date, review score only if submitted before the cut-off (otherwise a missing-value flag), and customer state or city.
Forbidden features	Anything dated after the prediction point: later orders, total number of orders, lifetime revenue, RFM values computed with later data, review scores submitted after the cut-off, and any field derived from the label.
Role of RFM	Full-history RFM is used for descriptive analysis and customer segmentation only. It is not a model feature, because it would contain information from after the prediction point.
Data split	Time-based split of first-order cohorts (earlier cohorts for training, later cohorts for testing), with stratified cross-validation inside the training set.
Preprocessing	Imputers, scalers, encoders and SMOTE are fitted on training data only and applied inside a pipeline, so no test information leaks into training.
Window check	The 90-day window is a starting value; it may be adjusted after exploring the data, and the change will be documented together with the reason.

3.6 Key Use Cases

Use case	Actor	Description	Result
UC-1 View business performance	Manager	Opens the Revenue and Customers views and filters by period	Sees revenue, AOV and customer trends

Use case	Actor	Description	Result
UC-2 Analyse products	Category manager	Opens the Products view and selects a category	Identifies top and weak products
UC-3 Target a segment	Marketing analyst	Opens the Segments view and reviews segment profiles	Selects a segment for a campaign
UC-4 Review predictions	Marketing analyst	Opens the Model Predictions view	Sees which customers are likely to repeat-purchase and why
UC-5 Request a prediction (optional)	Developer / system	Sends customer features to the FastAPI endpoint	Receives a repeat-purchase probability

3.7 Assumptions and Constraints

Assumptions

- A public, anonymised e-commerce dataset (proposed: Olist) represents the company's data and covers all five required entities.
- Customers are identified across orders by customer_unique_id, not by the per-order customer_id.
- The repeat-purchase target uses a 90-day follow-up window after the first delivered order (Section 3.5), which may be adjusted after data exploration.
- All team members have access to a laptop able to run Python, a SQL database and a dashboard tool.

Constraints

- Fixed deadline: the implementation must be finished before December 2026 (about four weeks of work in November).
- Team size of five members, each also managing other academic commitments.
- Only free or student-available tools may be used; no paid cloud services.
- Only one ML task is implemented, as required by the project brief.